

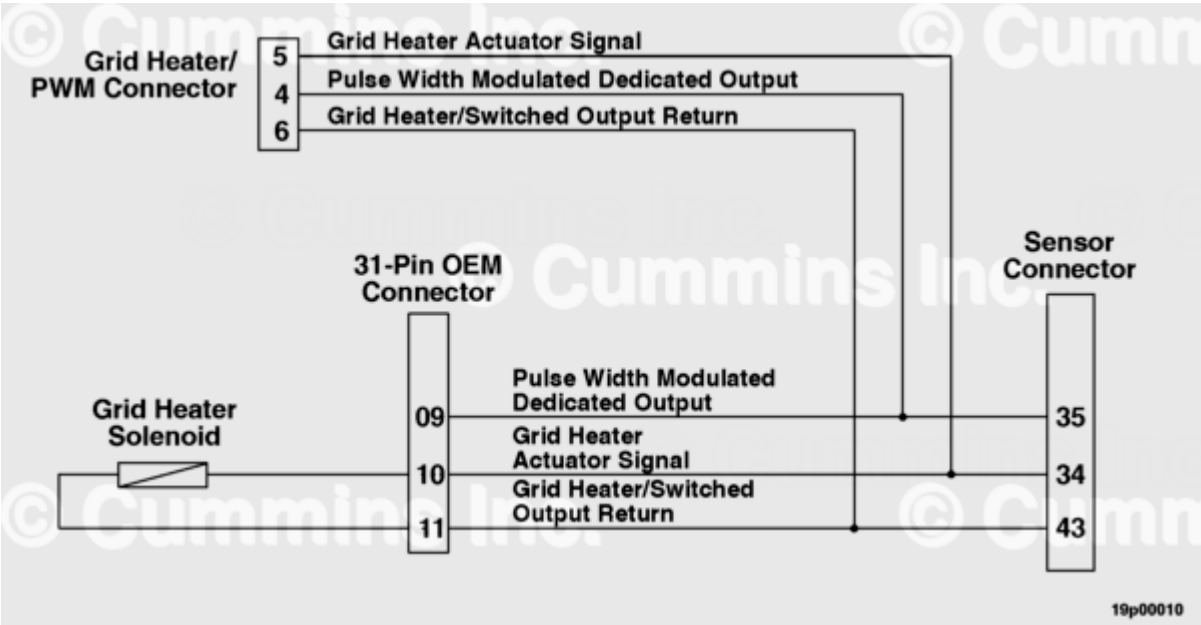
# Fault Code: 381

## Intake Air Heater Number 1 Circuit Error

 Printable Version

### Overview

| Codes                                                                                                     | Reason                                                       | Effect                                                                     |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|
| <p>Fault Code: 381</p> <p>PID(P): S237</p> <p>SPN: 626</p> <p>FMI: 11</p> <p>Lamp: Yellow</p> <p>SRT:</p> | Error detected in the cold start aid relay 1 enable circuit. | Intake air heater number 1 can be ON or OFF all of the time or is damaged. |



Intake Air Heater Number 1 Circuit

### Circuit Description

The intake air heater improves starting and white smoke control in cold ambient conditions. The electronic control module (ECM) controls relays that switch power to the air heater.

### Component Location

The intake air heater is located at the air inlet connection into the intake manifold. The location of the heater relays will vary with original equipment manufacturer (OEM).

## Conditions For Running The Diagnostics

This diagnostic runs when the keyswitch is in the ON position and when the device is activated. In some cases the diagnostic can also run at some fixed intervals.

## Conditions For Setting The Fault Codes

- Intake air heater circuits are short circuited.

## Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light immediately when the diagnostic runs and fails.
- The intake air heater operation will be disabled.

## Conditions For Clearing The Fault Code

The ECM will turn OFF the amber CHECK ENGINE light and the fault code state will become inactive immediately after the diagnostic runs and passes. The fault code can also be cleared with INSITE™ electronic service tool.

## Shoptalk

- These faults could also indicate the intake air heater circuits are shorted to battery positive (+). This would command the grids to be on full time. This will drain the batteries, burn out grids, and/or destroy intake gaskets.
- The intake air heater circuits will **only** activate when intake manifold temperature is below 19°C [66°F] when the keyswitch is in the ON position.
- If the intake air heaters are **not** installed and Fault Code 381 is active, then use INSITE™ to disable this feature, "Intake Air Heater."

Refer to Troubleshooting Fault Code t05-381. (</qs3/pubsys2/xml/en/procedures/82/82-t05-381.html>)